គណនាអាំងតេក្រាល

$$\int_0^1 e^{e^{e^{e^{e^x}}}} e^{e^{e^{e^x}}} e^{e^{e^x}} e^{e^x} e^x dx$$

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ដំណោះស្រាយ

$$\int_0^1 e^{e^{e^{e^{e^x}}}} e^{e^{e^{e^x}}} e^{e^{e^x}} e^{e^x} e^x dx$$

តាង $u = e^x \Longrightarrow du = e^x dx \quad \Longrightarrow dx = e^{-x} du$

$$= \int_1^e e^{e^{e^{e^u}}} e^{e^{e^u}} e^{e^u} e^u du$$

តាង $v = e^u \Longrightarrow dv = e^u du \quad \Longrightarrow du = e^{-u} dv$

$$= \int_e^{e^e} e^{e^{e^v}} e^{e^v} e^v dv$$

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តាង $t = e^{v} \Longrightarrow dt = e^{v}dv \qquad \Longrightarrow dv = e^{-v}dt$

$$= \int_{e^{e}}^{e^{e^{e}}} e^{e^{t}} e^{t} dt$$

តាង $\mathrm{l} = e^{t} \Longrightarrow dl = e^{t}dt \qquad \Longrightarrow dt = e^{-v}dl$

$$= \int_{e^{e^{e}}}^{e^{e^{e^{e}}}} e^{l}\, dl$$

គេបាន $I = [e^{l}]_{e^{e^{e}}}^{e^{e^{e^{e}}}} = e^{e^{e^{e^{e}}}} - e^{e^{e^{e}}}$

ដូចនេះ $\int_{0}^{1} e^{e^{e^{e^{e^{x}}}}} e^{e^{e^{e^{x}}}} e^{e^{e^{x}}} e^{e^{x}} e^{x} dx = e^{e^{e^{e^{e}}}} - e^{e^{e^{e}}}$

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